Statement of Work

[redacted]

In-situ mapping of defects and microstructure from weld pool thermal monitoring in wire arc additive manufacturing

Project Duration: 01/15/2025 to 12/31/2025

Statement of Work

This project's goal is to establish melt pool thermal imaging using an affordable, off-the-shelf camera as a qualified in-situ monitoring tool as defined in the NASA Technical Standard NASA-STD-6030 "Additive Manufacturing Requirements for Spaceflight Systems".

We will evaluate the possibilities and challenges associated with detecting the defects and mapping the solidification microstructure using the in-situ monitoring through the following objectives and related tasks to be performed at [redacted]

1. Use low-cost and off-the-shelf camera for measuring temperature and solidification conditions in the weld pool
    a. Task 1: Implement two-color thermography on WAAM machine at [redacted]
    b. Task 2: Analyze thermal images to determine solidification conditions
2. Correlate in-situ solidification conditions to defects and microstructure (Task 3 and Task 4)
    a. Task 3: Identify samples for ex-situ characterization of defects and microstructure and ship them to MSFC
    b. Task 4: Coordinate with MSFC on their characterization data and identify the onset of defects using the in-situ solidification conditions
3. Technology transfer: Install thermal imaging system on WAAM machine at NASA (Task 5).
    a. Task 5: Provide student support via NASA MSFC intern program and provide input on hardware and software framework to implement two-color thermography on MSFC's WAAM machine

We will primarily utilize the Lincoln Electric's wire arc AM equipment in [redacted] followed by replicating the monitoring setup at the MSFC's cold metal transfer wire arc AM equipment.

The material of interest is Alloy 718 for the following reasons: it is suitable for space applications; it is amenable for manufacturing on equipment at [redacted] and MSFC without major oxygen contamination; it has discernible solidification microstructure that correlates with the solidification conditions such as thermal gradient and solidification velocity, the product of which gives the solidification cooling rate; its reported defects (e.g. cracking) are observed to vary with the solidification microstructure and processing conditions.

While Alloy 718 is a suitable candidate for this initial demonstration and WAAM is the process of interest, the techniques developed in this project could be readily generalized to other alloys of interest with higher melting points as well as to other melting-based AM processes such as blown powder directed energy deposition and powder bed fusion.